

Run report

2026-10-04T041042Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	12	dataset	jets
seed	2,127,985,580	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	4.044×10^{-4}	best_epoch	87
compile_cache_warm	True		
variables	Mass and hard scale (IRC-safe kinematics): m [GeV], $\ln \rho$ Continuous angularities (IRC-safe jet shapes): $\lambda_{0.5}^1, w, \lambda_2^1$ Splitting and 2-prong substructure: $z_g, \tau_{21}^{(\beta=1)}$ Hadronization, multiplicity and fragmentation (IRC-unsafe): $M, n_{ch}, f_{ch}, p_T^D, q$		
mmd_sigmas_detector	$3.117 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.026 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
<i>RAN</i>			
data	1.429	4.5	cache hit – train
train	17.240	54.2	train
transfer	2.818		train
compile	5.177		train
epochs	6.226		train
select	2.645		train
particle_mmd	0.488	1.5	train
save	0.196	0.6	train
plots	6.727	21.2	train
evaluate	5.701	17.9	train
total	31.781	100.0	
<i>OmniFold baseline</i>			
parse_config	0.001	0.0	omnifold
data	0.840	0.2	cache hit – omnifold
omnifold	485.546	98.5	niter=3, epochs=50 – omnifold
init	2.184		DataLoaders and the two MLPs – omnifold
unfold	467.278		MultiFold.Unfold – omnifold
iter1_step1	133.033		detector-level reweighting – omnifold
iter2_step1	63.096		detector-level reweighting – omnifold
iter3_step1	82.694		detector-level reweighting – omnifold
iter1_step2	92.031		particle-level reweighting – omnifold
iter2_step2	51.677		particle-level reweighting – omnifold
iter3_step2	44.713		particle-level reweighting – omnifold
reweight	1.529		evaluating weights on z_target – omnifold
evaluate	6.448	1.3	12 variables – omnifold
total	492.835	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
m [GeV]	202.0	134.1	33.6	9.274	95.4	17.26	91.5
$\ln \rho$	230.2	21.69	90.6	17.82	92.3	23.62	89.7
$\lambda_{0.5}^1$	265.6	16.02	94.0	15.96	94.0	25.24	90.5
w	270.4	14.99	94.5	15.72	94.2	26.58	90.2
λ_2^1	262.4	17.02	93.5	17.95	93.2	30.32	88.4
$z_g^\dagger$	59.46	59.46	0.0	24.98	58.0	21.87	63.2
$\tau_{21}^{(\beta=1)}$	117.4	68.36	41.8	31.68	73.0	22.07	81.2
M	221.0	173.8	21.3	12.94	94.1	21.51	90.3
n_{ch}	218.4	170.9	21.7	17.13	92.2	24.25	88.9
f_{ch}	51.79	37.49	27.6	4.393	91.5	6.802	86.9
p_T^D	260.3	29.82	88.5	17.29	93.4	28.35	89.1
q	73.13	13.60	81.4	12.90	82.4	7.232	90.1

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
m [GeV]	6.646	3.869	41.8	0.2134	96.8	0.2760	95.8
$\ln \rho$	8.550	0.5419	93.7	0.2984	96.5	0.3588	95.8
$\lambda_{0.5}^1$	9.636	0.6629	93.1	0.3212	96.7	0.3529	96.3
w	9.431	0.5789	93.9	0.4145	95.6	0.4478	95.3
λ_2^1	8.284	0.5852	92.9	0.5682	93.1	0.5964	92.8
$z_g^\dagger$	0.7984	0.7984	0.0	0.2894	63.8	0.2674	66.5
$\tau_{21}^{(\beta=1)}$	3.412	2.091	38.7	0.4675	86.3	0.3956	88.4
M	9.559	7.587	20.6	0.3861	96.0	0.4358	95.4
n_{ch}	7.500	6.525	13.0	0.2579	96.6	0.2731	96.4
f_{ch}	1.115	0.8598	22.9	0.2518	77.4	0.2518	77.4
p_T^D	8.633	1.574	81.8	0.3783	95.6	0.4595	94.7
q	1.476	0.2967	79.9	0.1998	86.5	0.1893	87.2

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
m [GeV]	26.44	15.38	41.8	0.7976	97.0	1.012	96.2
$\ln \rho$	33.92	2.100	93.8	1.149	96.6	1.387	95.9
$\lambda_{0.5}^1$	38.25	2.629	93.1	1.266	96.7	1.392	96.4
w	37.43	2.278	93.9	1.609	95.7	1.749	95.3
λ_2^1	32.74	2.150	93.4	2.059	93.7	2.183	93.3
$z_g^\dagger$	3.186	3.186	0.0	1.157	63.7	1.069	66.4
$\tau_{21}^{(\beta=1)}$	13.58	8.324	38.7	1.849	86.4	1.561	88.5
M	37.86	30.06	20.6	1.429	96.2	1.635	95.7
n_{ch}	29.73	25.88	13.0	0.9817	96.7	1.041	96.5
f_{ch}	4.451	3.431	22.9	1.007	77.4	1.007	77.4
p_T^D	34.19	6.211	81.8	1.501	95.6	1.823	94.7
q	5.818	1.140	80.4	0.7552	87.0	0.7116	87.8

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	152.5	465.7	−205.4	16.94	88.9	19.18	87.4

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
m [GeV]	111.0	94.57	14.8	88.95	19.8	85.50	22.9
$\ln \rho$	258.8	29.51	88.6	8.786	96.6	10.74	95.8
$\lambda_{0.5}^1$	309.2	22.54	92.7	11.76	96.2	23.30	92.5
w	292.0	16.71	94.3	9.496	96.7	22.63	92.3
λ_2^1	264.3	18.09	93.2	12.29	95.3	25.72	90.3
$z_g^\dagger$	109.6	109.6	0.0	37.97	65.4	27.80	74.6
$\tau_{21}^{(\beta=1)}$	257.8	146.4	43.2	37.82	85.3	15.82	93.9
M	255.6	201.8	21.1	30.19	88.2	25.41	90.1
n_{ch}	234.2	189.4	19.1	29.59	87.4	24.28	89.6
f_{ch}	45.27	42.33	6.5	8.646	80.9	5.836	87.1
p_T^D	238.1	52.73	77.9	16.97	92.9	20.31	91.5
q	81.04	15.69	80.6	15.90	80.4	9.812	87.9

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
m [GeV]	2.207	11.87	−437.8	1.833	17.0	1.774	19.6
$\ln \rho$	10.52	1.644	84.4	0.2467	97.7	0.2678	97.5
$\lambda_{0.5}^1$	12.81	2.317	81.9	0.4628	96.4	0.5087	96.0
w	11.82	1.621	86.3	0.3800	96.8	0.4441	96.2
λ_2^1	9.357	1.060	88.7	0.4078	95.6	0.4501	95.2
$z_g^\dagger$	2.092	2.092	0.0	0.4473	78.6	0.3728	82.2
$\tau_{21}^{(\beta=1)}$	10.32	6.386	38.1	0.5971	94.2	0.3437	96.7
M	7.827	12.52	−60.0	0.5071	93.5	0.4666	94.0
n_{ch}	6.567	10.22	−55.7	0.4487	93.2	0.3802	94.2
f_{ch}	1.214	1.061	12.6	0.3205	73.6	0.3194	73.7
p_T^D	8.813	4.183	52.5	0.5649	93.6	0.5907	93.3
q	2.260	0.8436	62.7	0.2305	89.8	0.2154	90.5

VLC divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
m [GeV]	8.707	45.68	−424.6	7.216	17.1	6.939	20.3
$\ln \rho$	41.74	6.496	84.4	0.9266	97.8	1.011	97.6
$\lambda_{0.5}^1$	50.79	9.226	81.8	1.826	96.4	2.007	96.0
w	46.87	6.421	86.3	1.468	96.9	1.724	96.3
λ_2^1	37.09	4.173	88.7	1.568	95.8	1.740	95.3
$z_g^\dagger$	8.289	8.289	0.0	1.788	78.4	1.490	82.0
$\tau_{21}^{(\beta=1)}$	41.02	25.40	38.1	2.371	94.2	1.360	96.7
M	30.97	48.60	−56.9	1.935	93.8	1.774	94.3
n_{ch}	26.00	40.04	−54.0	1.708	93.4	1.434	94.5
f_{ch}	4.846	4.229	12.7	1.280	73.6	1.276	73.7
p_T^D	34.77	16.58	52.3	2.219	93.6	2.313	93.3
q	8.835	3.281	62.9	0.8621	90.2	0.7853	91.1

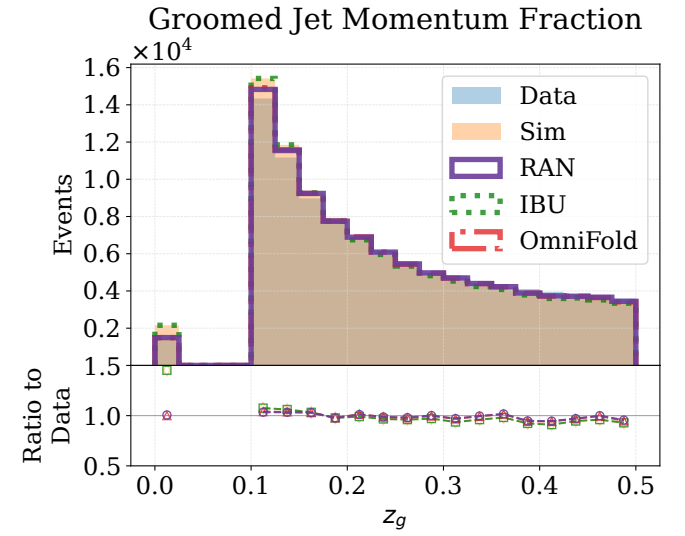
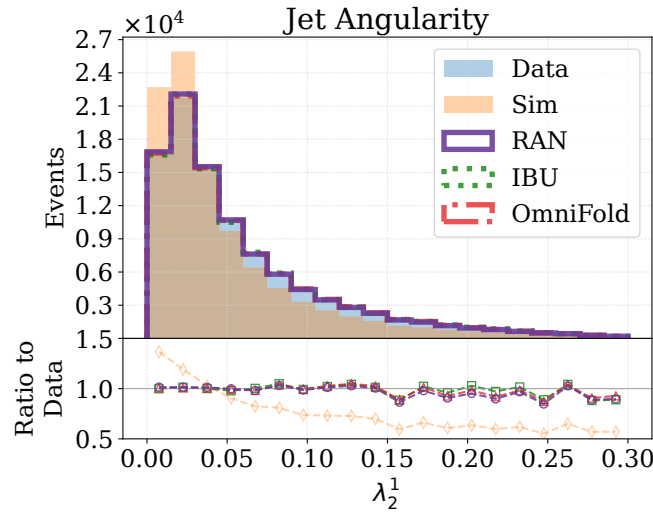
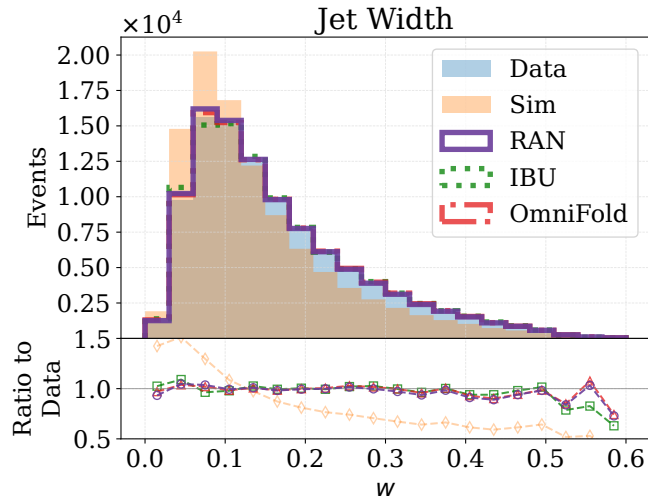
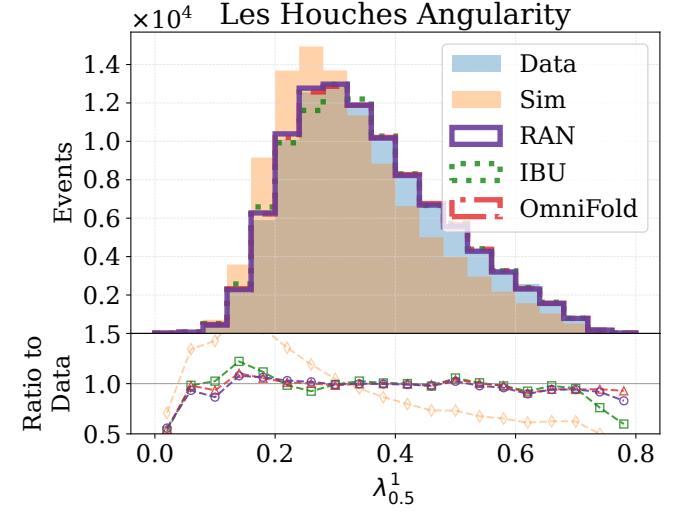
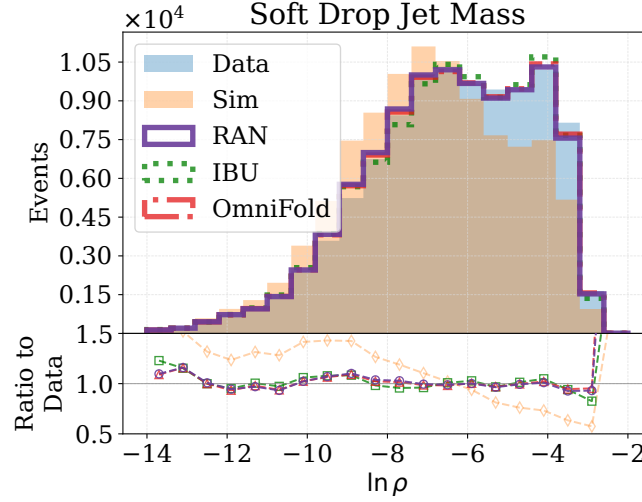
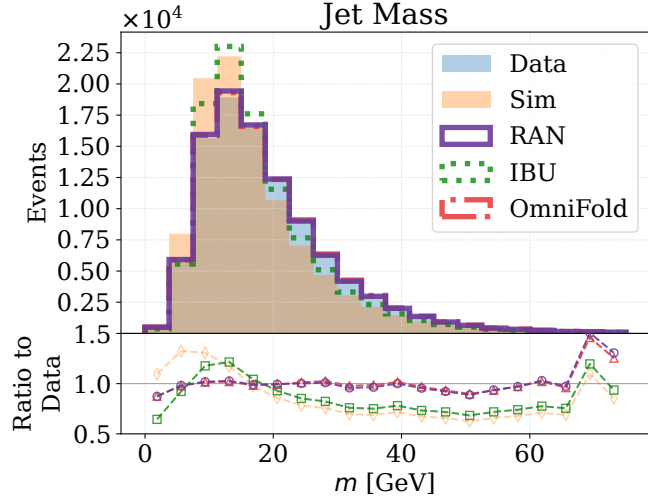
[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

Sliced Wasserstein, joint $[\times 10^3]$

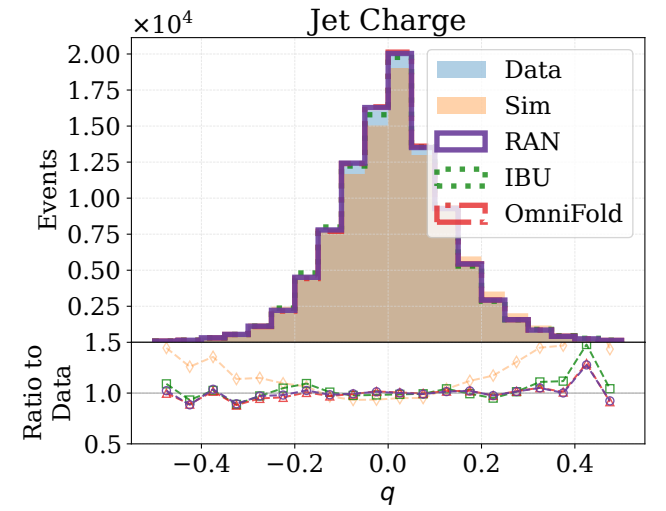
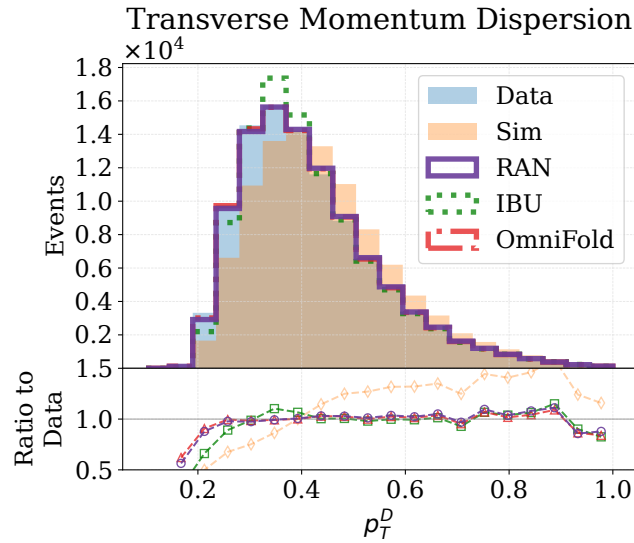
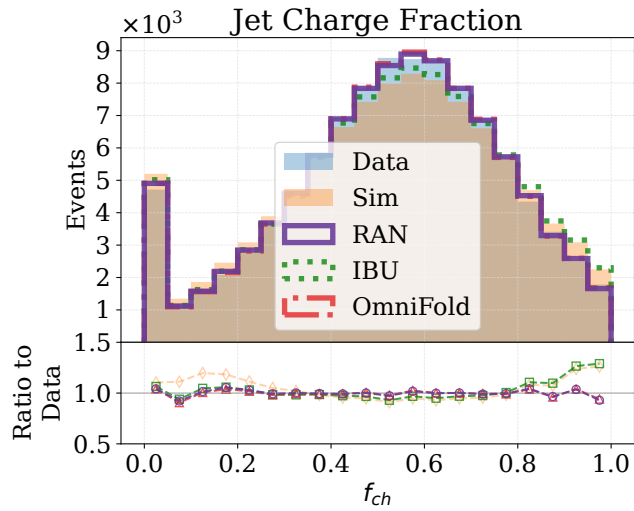
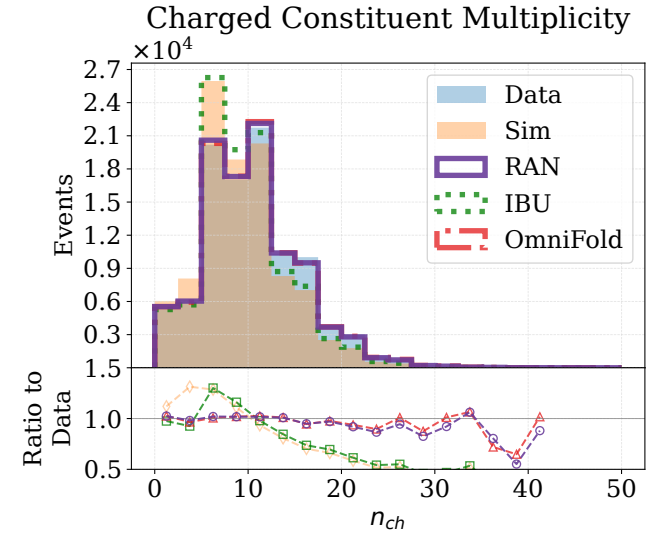
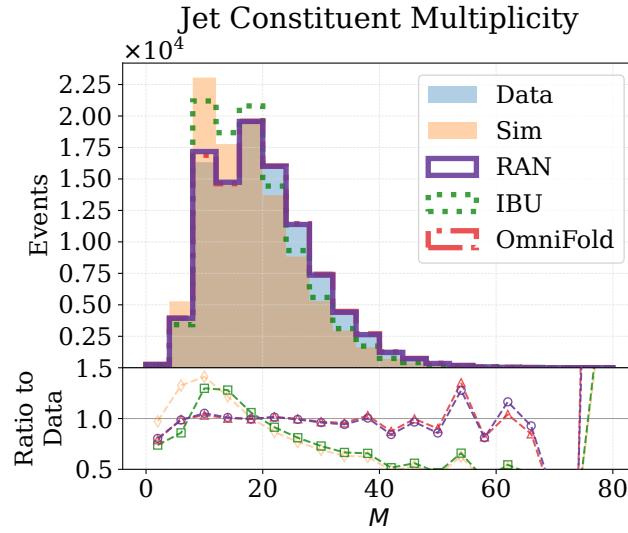
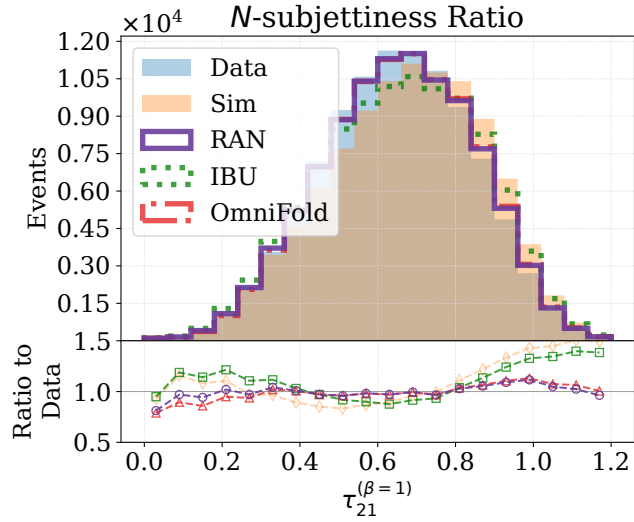
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
<i>All</i>	175.9	543.2	−208.8	27.95	84.1	25.25	85.6

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

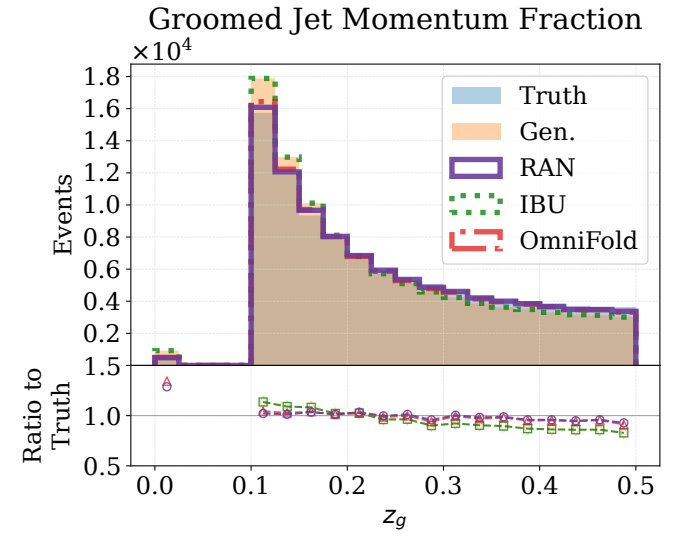
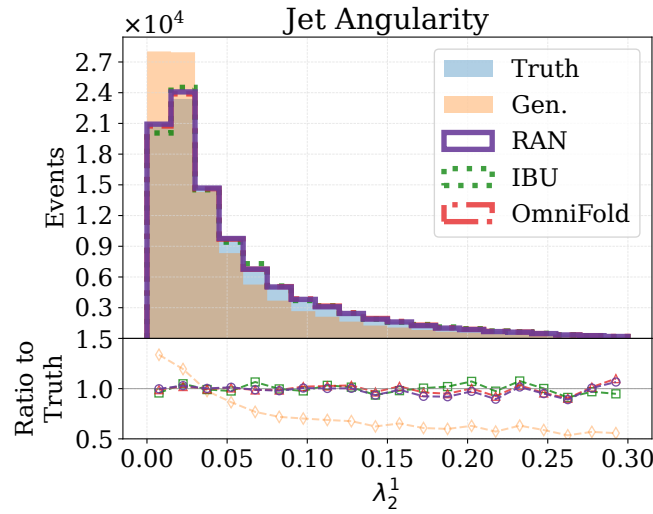
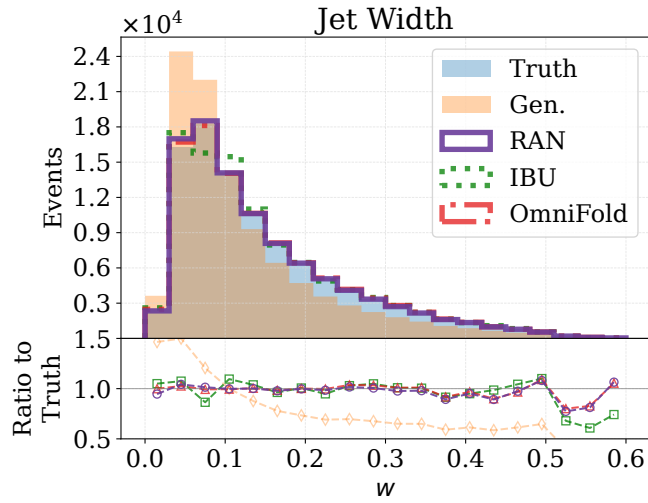
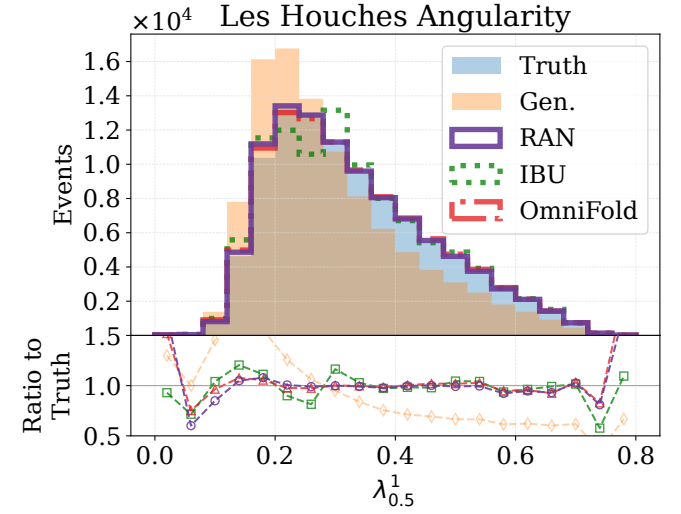
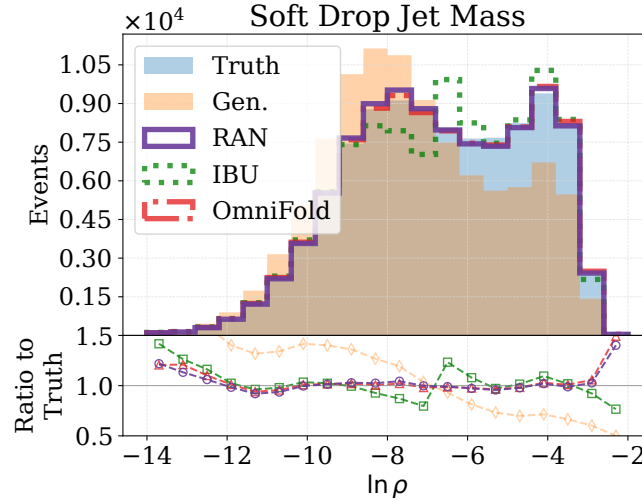
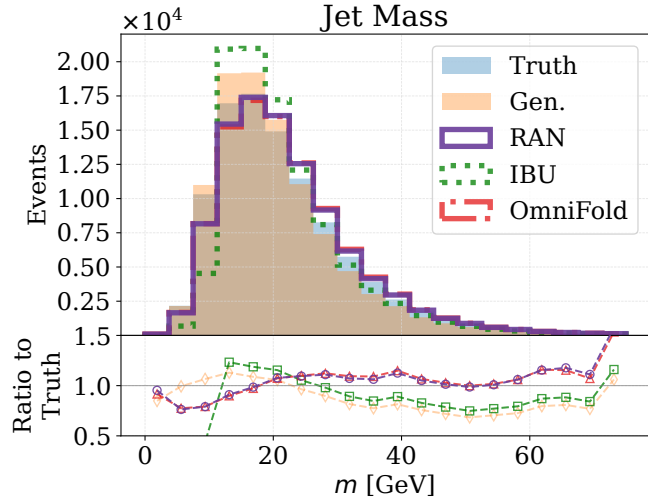
Detector Level (1 of 2)



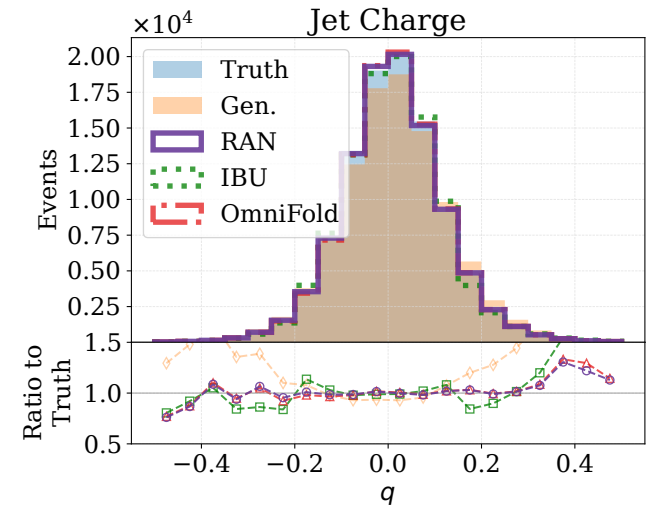
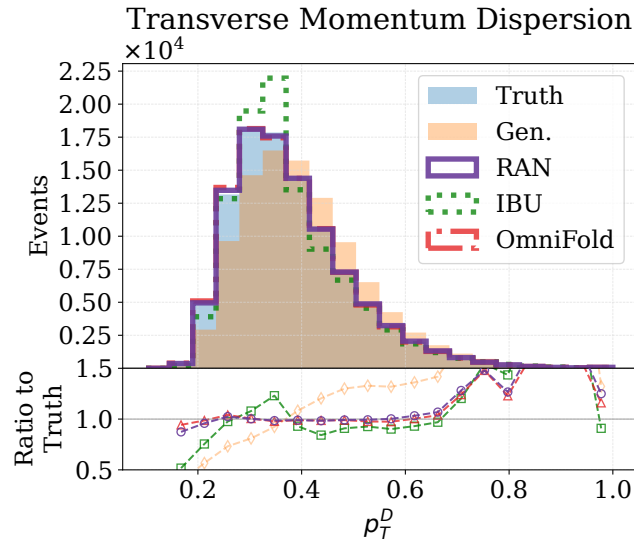
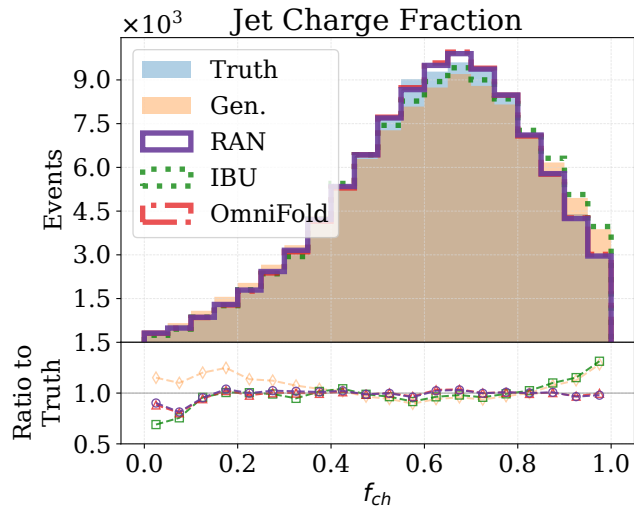
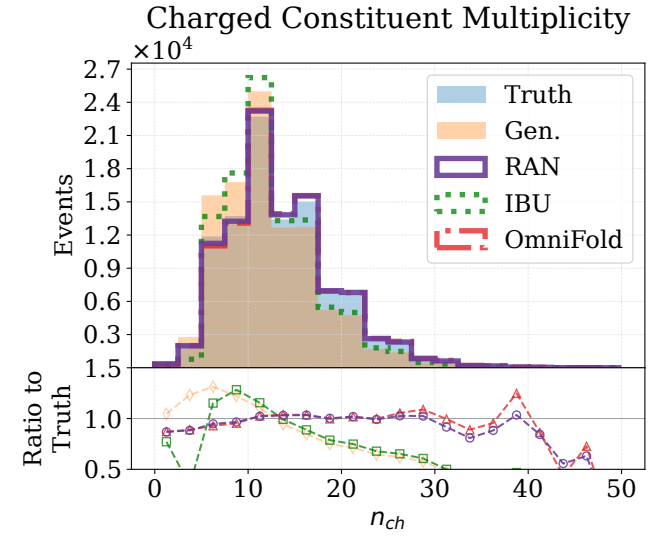
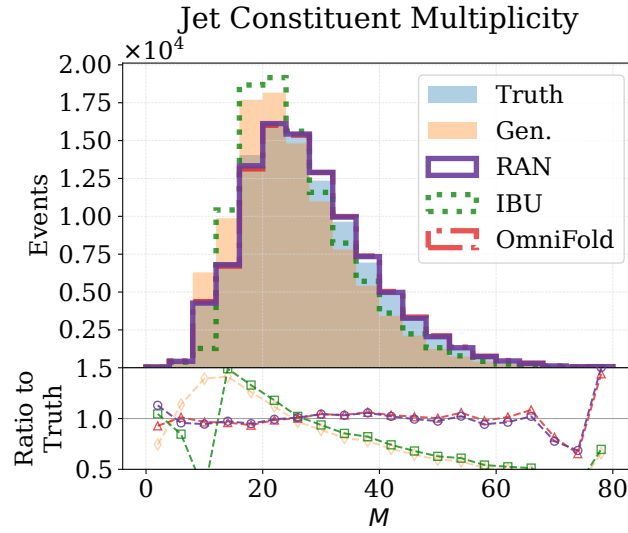
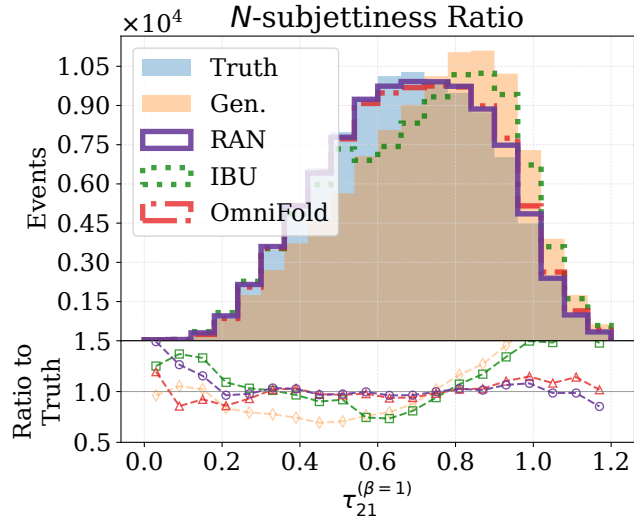
Detector Level (2 of 2)



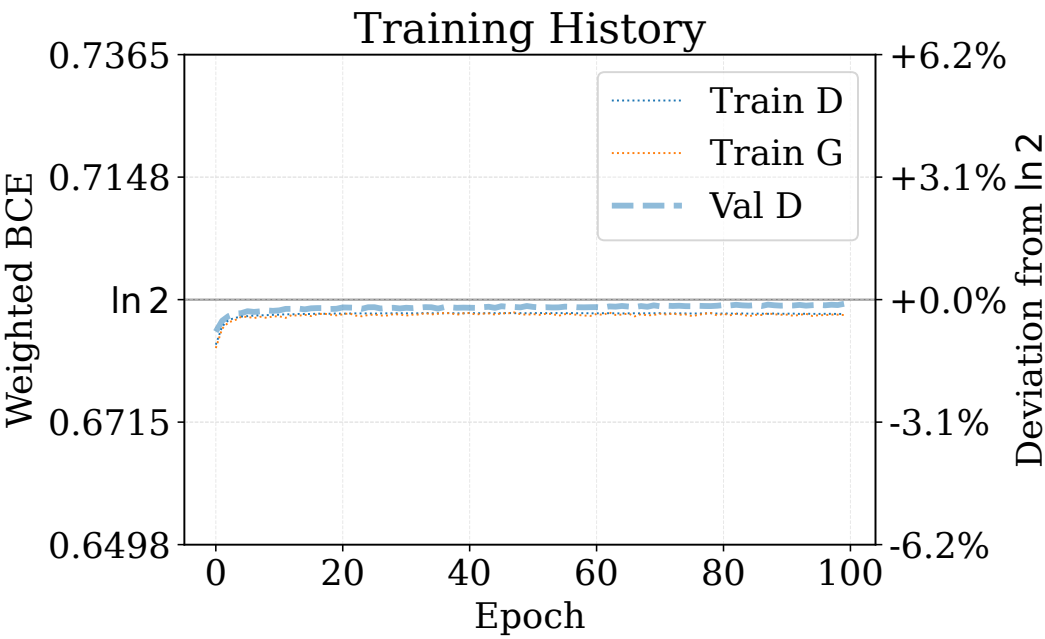
Particle Level (1 of 2)



Particle Level (2 of 2)



Training diagnostics
Training losses



Selection criteria

